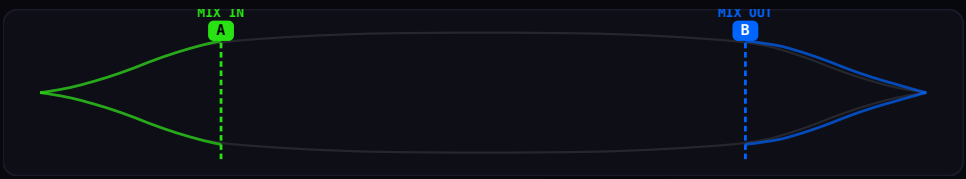
● OPEN SOURCE DJ TOOL

Playlist Convertor

Paste a **Spotify playlist URL**, get a folder of **DJ-ready MP3s** with BPM, key, Camelot codes, mix point hot cues, and a Rekordbox XML — plug in your USB and play.

Spotify URL → Download → Analyze → Tag & Cue → Export



v0.1.0

FEATURES

Everything a DJ Needs

From Spotify to the booth. Every step automated, every format supported.

BPM Detection

librosa beat tracking with auto-normalization to the DJ range (60–180 BPM). Halves or doubles to avoid false readings from complex rhythms.

Key Detection

Krumhansl-Schmuckler algorithm over CQT chroma features. Returns key name + confidence score for all 24 major/minor keys.

Camelot Wheel

Auto-mapping to Camelot notation (1A–12B). Compatible key lookup for harmonic mixing — same number, ±1, or relative major/minor.

Mix Point Detection

Energy-based analysis finds where the intro ends and outro begins. Beat-snapped, confidence-scored, written as Rekordbox hot cues.

Rekordbox XML

Full collection + playlist XML with BPM, key, Tonality, and POSITION_MARK hot cues. Import once, sync to USB, ready for CDJs.

Serato Compatible

ID3v2.3 tags, CBR validation, genre separator checks. Files pass Serato's format scanner without manual fixing.

DJ Set Builder

Build sets from your library. Filter by BPM range, Camelot key (with harmonic neighbors), playlist name, or free-text search.

FAT32 Safe

Filenames stripped of illegal chars, accents transliterated via Unidecode, paths capped at 255 chars. Works on every CDJ and controller.

TECH STACK

Under the Hood

Python CLI. Battle-tested audio libraries. No cloud dependencies.

`Python 3.10+` `spotdl + yt-dlp` `ffmpeg` `librosa` `mutagen` `Click` `Rich` `Unidecode` `NumPy`

PIPELINE

How It Works

Six stages from URL to USB. Parallel downloads, sequential analysis, one XML at the end.

1. **Fetch Playlist Metadata**
   Scrapes Spotify's embed page to extract track titles, artists, URIs, durations and artwork. No API key needed — bypasses restricted Development Mode.

2. **Download Audio**
   4 parallel workers via `spotdl` (wraps `yt-dlp` + `ffmpeg`). Each track matched on YouTube, converted to MP3. Skips tracks already in library.

3. **Analyze BPM, Key & Mix Points**
   librosa loads at 22050 Hz mono. Beat tracking for BPM, CQT chroma + Krumhansl-Schmuckler for key, RMS energy for mix-in/mix-out points.

4. **Tag MP3s**
   ID3v2.3 tags: title, artist, album, cover art, BPM, key, Camelot, and mix points as `TXXX` frames.

5. **Export Rekordbox XML**
   Collection + playlist XML with `POSITION_MARK` hot cues. Green A = mix in, blue B = mix out. Import and sync to USB.

6. **Organize & Archive**
   Files renamed with track numbers, folder gets BPM range. Tracked in `.library.json` for dedup and set building.

USAGE

Two Ways to Run

Wizard

```
$ pconv

Spotify playlist URL:
https://open.spotify.com/playlist/3c...

Target DJ software: 1 (Rekordbox)
Analyze BPM & key? Yes
Detect mix points? Yes

┌─ Ready to download ──────┐
│ Analyze: yes + mix points│
└──────────────────────────┘
```

CLI

```
# Quick download
$ pconv download URL

# Full analysis + mix points
$ pconv download --analyze-mix-points URL

# Set builder
$ pconv set new
$ pconv set add --bpm 120-128
$ pconv set add --key 8A
$ pconv set save
```

OUTPUT

What You Get

One folder, ready for USB. Files, XML, and library tracking.

```
output/

output/
└── 2026-02-28_Luca_rekordbox_120-128bpm/
    ├── 01 - CKay, Olamide - WAHALA.mp3
    ├── 02 - SPINALL, Fireboy DML - Sere.mp3
    ├── 03 - Burna Boy - Last Last.mp3
    ├── ...
    └── Luca.xml ← Rekordbox import file
```

#	TITLE	BPM	KEY	CAMELOT	MIX IN	MIX OUT
01	WAHALA	124	Ab minor	**1A**	0:32	3:05
02	Sere	120	Eb major	**3B**	0:16	3:22
03	Last Last	128	F minor	**4A**	0:24	2:48
04	Free Mind	122	Bb major	**6B**	0:28	3:40

MIX POINTS

Hot Cues on Your CDJs

Import the XML and mix points appear as color-coded hot cues, beat-snapped and named.

A	B	C	D	E	F	G	H
Mix In	Mix Out	Empty	Empty	Empty	Empty	Empty	Empty
0:32	3:05	–	–	–	–	–	–

CONFIGURATION

Mix Point Tuning

All constants in `config.py`. Adjust to match your genre and mixing style.

ENERGY_THRESHOLD `0.70`

Fraction of body median energy to trigger. Lower for ambient, raise for EDM.

SUSTAIN_BEATS `8`

Consecutive beats above threshold to confirm a crossing point.

SMOOTHING `4.0s`

RMS smoothing window. Larger ignores short breaks in energy.

MIN_DURATION `60s`

Tracks under this length are skipped. No mix points on interludes.

FALLBACK `16 beats`

Default intro/outro when energy is flat or detection fails.

POSITION `33/67%`

Mix-in clamped to first third, mix-out to last third of track.